Bose-Einstein Condensate, Collisions, The Physics of Fire â€“TYTYK

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis: This episode explores fundamental scientific principles that often contradict classical intuition, specifically focusing on quantum matter states (Bose-Einstein Condensate), the application of kinetic energy laws to vehicular safety, and the chemical requirements of combustion. The hosts argue that understanding these phenomena requires moving beyond human sensory experience (classical intuition) to apply rigorous quantum mechanics or thermodynamic conservation laws, ultimately emphasizing practical applications like improved safety engineering and sustainable carbon cycles.

2. Key Arguments & Data:

    - **Bose-Einstein Condensate (BEC):**
        - **Definition:** A phase of matter where particles are cooled to fractions of a Kelvin (near absolute zero) so their wave-like nature dominates, causing them to overlap into a single "metaparticle" system.
        - **Quantum Behavior:** Particles do not function as discrete objects but as overlapping waves; cooling reduces vibration and increases wavelength until resonance occurs.
        - **Heisenberg Uncertainty Principle:** Applied metaphorically to explain position vs. energy knowledge, though the primary focus is on wave function overlap.
        - **Applications:** Superfluidity (zero fluid friction) and superconductivity (zero electrical resistance), leading to technologies like maglev trains and quantum computing.
    - **Kinetic Energy in Collisions:**
        - **Energy Conservation:** The objective of a crash is not to stop motion but to absorb kinetic energy ($E \propto v^2$) without transferring it to the passenger.
        - **Velocity Scaling:** Doubling speed quadruples kinetic energy; a 60 mph impact has four times the energy of a 30 mph impact.

        - **Deceleration Time:** Injury is caused by rapid deceleration (high negative acceleration). Safety devices (seatbelts, airbags) act as "marshmallows" or crumple zones to extend the time interval over which energy is absorbed.
        - **Case Studies:** Usain Bolt running into a wall at 25 mph (high injury risk); Dale Earnhardt's death due to rapid internal organ deceleration; High jumpers landing on foam pads vs. cement.
    - **Combustion & Thermodynamics:**
        - **Oxygen Requirement:** Burning requires molecular interaction with oxygen; substances like water or aluminum lack this structure and do not burn, despite high heat capacity.
        - **Carbon Dependency:** Burning is a chemical reaction breaking carbon-based molecular bonds. Almost all food and fuel contain carbon atoms (except salt/water).
        - **Hydration Effect:** Wet wood is difficult to burn because internal water must be vaporized first, absorbing energy and preventing ignition until steam expands.
        - **Soot vs. CO2:** Incomplete combustion leaves solid carbon ash (soot); complete combustion produces carbon dioxide gas (CO_2).
        - **Carbon Neutrality:** Burning wood returns the CO_2 that trees previously absorbed, making the process neutral provided replanting occurs.

3. Actionable Takeaways:

    - **Adapt Safety Physics:** Prioritize vehicles with "crumple zones" and airbags designed to maximize deceleration time rather than stopping a collision instantly. Avoid scenarios where passenger space is compromised by rapid impact.
    - **Understand Velocity Risk:** Recognize that increasing vehicle speed exponentially increases risk; the kinetic energy increase is quadratic, not linear.
    - **Fuel Selection for Fire:** Ensure firewood is thoroughly dry to remove internal moisture before combustion begins, as water

inhibits burning until it evaporates.

- **Embrace Quantum Reality:** In research and education contexts, acknowledge that quantum phenomena cannot be predicted by classical intuition (e.g., matter behaving as a single wave) and requires specific data-driven understanding of absolute temperature scales rather than "feeling" cold.

Full Transcript

Hello, Starktalkians. Neil deGrasse Tyson here. Coming up is another installment of Things You Thought You Knew, in conversation with Chuck Nice. First, we talk about the Bose-Einstein condensate. What is that? You'll find out. After that, we talk about the kinetic energy of car collisions. And then we round it out with a discussion on burning things. Check it out. Welcome to Star Talk, your place in the universe where science and pop culture collide. Star Talk begins right now. Chuck, I've been itching to do this for so long. Uh-oh. Okay. This is going to be on the Bose-Einstein condensate. Okay, okay. And I bet you've been just itching to know about that. Oh my God, I can't tell you how many sleepless nights I have had.

Thinking about the Bose-Einstein condensate. I'm actually building a particle collider in my basement, because it just means that much to me. Anyway, I don't even know what is the Bose. Well, you've heard of Einstein. Bose was an Indian physicist. And the boson is a type of particle that is named in his honor. And you might have heard of the Higgs boson. It's a different kind of boson. Here's what's interesting with a Bose-Einstein condensate. It's it's a it's a subdivision of condensed matter physics where matter gets very dense and very close to itself and you can do that by reducing the temperature so that the particles don't vibrate as much. Gotcha. Okay, and as you make it colder and colder and colder and colder. All right. Now here's what's going on. Yeah It needs a sweater. And I have to whisper because it might disturb the particles, okay? It's a beautiful day here at Augusta. Winds are slightly out of the west as the particles line up on the 18th and we've got a... By the way, something about sweaters, if you're the same temperature as the outside air, a sweater will do nothing. It can't warm you. Wait, if you... Oh my God, yes, you're right.

Yes. A sweater is not inherently warm. The sweater is just fabric that's an insulator, really. Correct. So if you are warmer than the outside, and then you put on a sweater, it keeps your warmth in you. Right. But if you are the same temperature. Same temperature as the other. Right. As you just said, your clothing won't do anything. By the way, I do know, and I hate to say it and don't write, I know a couple of women who will put on a sweater when

it's 98.6 degrees. Really? Why is it so chilly in here? Oh my God, it's so cold. The thermodynamics of men and women have never been the same. You know they did a study and they found that women are actually colder. No, they did a study and found out that the temperatures in office buildings were specked for men because it was at a time when only men were occupying the office buildings. And so yeah, the women end up colder. Had they speculated for women, men would be complaining how hot it was. Okay, you're right. You know, that's why I'm glad I'm a man. No, I'm joking. Do not write. That is not, that's called a joke. That's it. Okay, do not write.

Okay, that would have rocked the house in 1987. It would have, right. You know, let's get back. So these particles, you're lowering the temperature. Okay. So in quantum physics, every particle is also a wave. Okay. Holy Jesus. Give me a second, man. Give me a second. Quantum physics, every particle is also a wave. Now, is that because they're vibrating? You want to ask rational questions about a reality that still transcends our common sense. Okay. Got you. All right. So what you're saying here is like, these are things you have to accept by suspending. your natural understanding, or your understanding of your own. Your limited understanding brought to you by your own five senses, honed on the plains of the Serengeti to not be eaten by a lion. The universe is vastly more complex than that, and involves statements that on the face of it seem like they can't be true, but they can't be true.

Because your intuition is not quantum intuition, it's classical intuition. That's what I mean by your natural understanding, it's the world around you. So now, let's get back to this. Every particle is a wave on the quantum level. Correct. And so the quantum is also a wave. So you have a jumble of particles and they're all doing their thing. You can look at what wave is associated with them, what wave length. And they're each doing their thing. They're all independent particles. Okay. So let's cool them down. As they vibrate less, they have less energy associated with them and their wavelength grows. Okay. Okay. All right. So just hang with me now. I don't want to lose you. Okay. I'm with you right now. Right now. Okay. Okay. The characteristic wave of the particle as it slows down, begins to grow. Now there's what's called the Heisenberg uncertainty principle, which is the more you know about a particle's position in space, the less you know about

its energy, and the more you know about its energy, the less you know about its position in space.

Okay, now there's a geeky way to get out of a speeding ticket Where the cop stops you and It says, do you know how fast you were going and I and you just say no That means I'm everywhere or something. There's some stupid quantum Okay, that's surely to get the black man shot and then There you go. That's how that works. That's how that works. All right, so anyhow So you have these particles and as You cool this material when I say cold I don't mean put it down in the freezer Yeah, I'm talking on like Kelvin absolute temperature scale where zero is the coldest we have a whole explainer on that one Yes, we do I remember yes, you can get down to ten degrees and nine eight seven six five four three two one a half a degree a tenth of degree a thousandth of a degree. You just keep getting this colder and colder and colder and the wavelengths of each of these particles gets longer and longer. And you reach a point where all the wavelengths overlap each other for every single particle. And they become one. They've condensed. They've condensed and become one.

functioning particle. Oh! Nice. Okay, they didn't just condense and become a solid thing. No, all their waves, which used to be separate and distinct, are now overlapping and resonating with each other. They're the Voltron of particles. Okay, I forgot who Voltron is. Who's that? That's the one where all the little robots come together to make one big giant robot. Yes, there it is. There it is. What movie was that? What comic book was that? Voltron. Well, Voltron is the story. Yeah, that's the story. OK, that is it. Yeah. So all the particles come together and they become one metaparticle. A functioning metaparticle. Correct. Because they all resonate together at the same time. And this is why you can't think of matter as only particles or waves, all right? Because depending on the situation, they will function as one or the other, and in some cases, both. Wow. And so, oh yeah. And by the way, there's a lot of phenomena. So that's a Bose-Einstein condensate. It happens like quick. As you drop the temperature, it transitions, boom! It's basically a phase change in what it is. It's weird for us to think that.

But is it really any weirder than having a transparent liquid all of a sudden become solid? Right, not really. Think about it. If you've never been in a

freezing temperature and someone has a... A cube of ice. A cube of ice and you say, what's that? That's water. And they say, no, it's not because I can't drink it and it's not a liquid. And it's not... Rich! Rich! Why do they only burn the witches and not the warlocks? You know, why did they never find the warlock is the question. So men had all the power, I guess. Which I don't understand why every woman, the person who ever, because it's always a guy, right? You know, witch. And immediately, if I was the woman, I'd have been like, yes, I'm a witch. He's my warlock.

Yeah, totally take that one out. We're going down the gutter, baby. Fight the fire with fire. Fight the fire with fire. So, other phenomena like this, one of them is called super fluidity. You know what that is? The fluid becomes a super fluid, which means if you pass something through it, like your fist, there is no resistance to your hand moving through that volume. It just envelops your hand as though nothing is there. It's super fluidity. Then you can have superconductivity where the material offers no resistance to the movement of electricity. So you don't lose any electric power to heat. Okay? And you have perfect transmission of electricity. And so you can have superconductivity and from that you can get superconducting magnets, which are extremely powerful.

I'm still meditating. Yeah, you get some of that. You get some of that. Okay. Well, you get a maglev train where that would help out. It could be totally happening if you can get superconducting magnets to do that. The problem is you can't cool everything to fractions of degrees above Kelvin. You need sort of high temperature superconductors, high temperature, which would still be very cold. And so it's a whole thing that it's a whole frontier in material science and in physics. Wow. So the point is none of these phenomena make any sense in a classical world. But when you bring in quantum forces and put it under those conditions, you need quantum physics to understand it. And so I just think it's cool that all these particles can condense and act as one meta system. And you said it's not Ultron. Who was it? Did you say? Voltron. Voltron is a some of smaller robots that's acting as a single robot. That's a single robot. Yeah. It's Voltron is a Bose-Einstein condensate. Look at that. That's pretty awesome. Yeah. That's right. So there it is. That was quick. No, it was great. Yeah. We need a healthy respect for quantum physics as we enter the realm of quantum computing,

where once again we exploit the weird wacky stuff that is normal in quantum physics and is out of our life experience.

from our senses. And so there it is, Chuck. Okay. That is super duper fly. No, it's super duper conducting. Oh, by the way, speaking of super duper, you're going to build the world's largest particle accelerator. And you know how you accelerate particles? Okay, you get a particle with a charge in it, and then you shoot it into a magnetic field. Okay. And if you vary that magnetic field, you can accelerate the particle, but the particle wants to have its direction of motion changed. It starts arcing. So you know exactly how much it's going to arc. You make a tube that's round, and then you alter the magnetic field in that tube to accelerate the particle. This is how every single circular particle accelerator works. And you can only accelerate particles that have charges, can't accelerate neutrons because they don't have charge. That's that joke in the bar.

Neutron walks into a bar and says, I like a beer. So here it is. And the neutron says, how much? And the bartender says, for you, no charge. That's stupid. So just as we go forward, just embrace all that quantum physics brings you and don't try to question it. Yeah, you can't. You can't because you're questioning it from a foundation that is not the world in which of laws that govern quantum phenomena. So you just have to grow accustomed to quantum physics. If anyone says, I understand it now. No, you don't. Right. No, no, it just is. Right. It just is. That's great. All right, dude. That was amazing. Yeah. Weird wacky things on the frontier of quantum physics and what it does. Weird wacky things. Yes. That was Johnny Carson. That's correct. You are correction. Here's yet another fact from Lost in Space.

the latest collaboration between StarTalk and National Geographic. 1995, we discover the first authentic exoplanet, a planet orbiting another living star. So, anyone born from 1995 onward? I now declare each one of you to be Generation X-Oplanet. If you like that fact, you can find 4,000 999 more in Lost in Space. 5,000 facts to help navigate the universe. Lost in Space is now available wherever you get your books. This one is on safe car accidents. Okay, I'm in the oxymoron, so I'll play along. Okay. A safe car accident. A safe? Car accident. Correct. Okay. So let's look at the physics of

a moving vehicle. All right. So let's assume you're wearing your seatbelt. So you're attached to the car. All right. You have all this stuff up front, you know, under the hood, the engine. If it's a front, you know, front wheel, whatever, front engine car, you have stuff in behind you. You got the doors to your left and right. And nowadays we have airbags.

Correct. In all those places. The doors have it. Side curtain airbags. Side curtain airbags. Right. Yeah. Okay. So the only objective in a car accident is to take the kinetic energy of the car and do something else with it. Okay. In a way that does not harm the passenger. The passenger. Okay. All right. So this is why you might have heard these things about crumple zone. You say, why would I want to crumple my own damn car? Because the energy of the speed of the car now goes into crumpling your car because your car after the accident is going zero miles per hour. Where did that kinetic energy go? It has to go something. Something's got to break. You don't want it going in your head. In your head, OK? Something's got to break. OK. So if you're not wearing a seatbelt, And you're going 60 miles an hour and then zero. And if that happens relatively quickly as what happens in car accidents. You become a bug. You keep moving. So what happens is the car hits the wall and the car stops moving. Right. But you're not attached to the car. You keep moving at what speed? 60 miles an hour. 60 miles per hour into the windshield. Correct. Now there's some people who don't wear seatbelts below like 30 miles an hour. Okay. So I'm not going fast enough. I'll wait till. Okay. Here's, here's, here's a thought experiment. Yeah. Do you know how fast Usain Bolt was running when he set the world record? No. In the hundred meters, about 25 miles an hour. Okay. Wow. Maybe 24, something like that. All right. So ask him, uh, who's saying, please do this experiment. See that brick wall run as fast as you can face first into that wall. And then come back and tell me about how, tell me how that went.

Okay, so he would break his face probably die Yeah, yeah, and that's that's 25 miles an hour. Yeah, but at least he went out on top You know My my favorite Usain Bolt plaque. Did I tell you this? No, I was someone I was somewhere in Europe I don't remember it might have been Heathrow one of the airports and There's a plaque there this after he said his world record in the Olympics. There's a plaque on the ground and says Right here in, you know, in 2000 and whatever, whatever was his Olympics 2012. Usain Bolt

arrived at Heathrow Airport. His plane arrived 10 minutes later. That is amazing. That is a really cute plaque. Oh, man. Oh, yeah. That was, that was good. Okay. I mean, clearly you are the fastest human, period. You're the fast. If that's, if they give you a plaque, this is a plaque.

A plaque that's fun, a fun plaque and plaques are always dry and interesting. This was a fun plaque. But that's, that's, that's like Muhammad Ali. He was like, I'm so fast. I'm so fast. I turn out the light and I'm in the bed before the room gets done. So you're going 60 miles an hour. Of course, the car has stopped and now boom. Now you go 60 miles an hour and energy goes as the square of the velocity. So you have four times the energy. as someone going 30 miles an hour. And you don't want to go 30 miles an hour because just imagine Usain Bolt running into a brick wall that way. All right. So the whole thing here is you want to reduce the time it takes you to go from 60 miles an hour to zero. Right. Okay. That's deceleration, a negative acceleration. Did I say reduce it? Excuse me. You want to increase the time. Right. Because the faster that happens, the more damage that will be to your body.

Okay, so when Dale Earnhardt was in an accident in a track and hit some embankment on the side, I think it was a one car, I think there was no flames and it just stopped. I say he's dead. Because his body is going 200 miles an hour and a fraction of a second later he's going zero. That's the kind of thing. Well, is his body strapped in? Sure. But is his head attached? Okay. Your head is going 200 miles an hour. And it's just attached with some ligaments and tissues and some bones. And that'll rip off. Your heart is dangling there in the middle of your chest in one of its arteries. It's going 200 miles an hour. And in a fraction of a second, it's going zero. Your heart punch basically can get ripped out of your chest and not come out of your body necessarily, but rip off of the part that's holding it.

holding it up. So things can happen internal damage that'll just kill you. Kill you. Okay. So what you want to do is you want to stretch out as long as possible the time it takes the car to slow down. Okay. That's what you want to do. And while that's happening, have the car bear the brunt of the reduced kinetic energy. Let the car get banged up, beat up. Let that happen to the car. Let's just make the cars out of marshmallows. Thank you. Okay. And

that's kind of what airbags are, are marshmallows. Okay. Right. So not as delicious though. Okay. So when you do that, you reduce the stress on the body, giving your body a chance to slow down with the car.

in a way that does not bring bodily harm to you. OK. So there's a video on YouTube. Some years ago, since I have no reason to think it's not still there, called And They Walked. Oh, I get it. This is a video of the most twisted, fiery car wrecks in NASCAR and Formula One history. And in every one of these wrecks, the person just steps out and walks away while the fire folks come with the hoses. And they say, and they walk. And they say, it's a miracle. No, it's not. It's physics. Because every one of those wrecks, the car went a quarter mile down the track. spinning, twisting, bending. And there you are inside of it. That's quite the hay ride you're on, okay? But what kills you is that happening quickly, not happening slowly. So the longer it takes, the more twisted the wreck is for how long it took to come to a stop. And by the way, they're all in roll cages anyway, okay? All right, so let's get that straight. So they're all in a protective cage.

The tires fly, flames come out, plus they're in flame-proof suits. So what we're really talking about here are you gonna break your bones. So the longer it takes the accident to slow down to a stop, the safer you are, period. And if you roll your car and if your hood doesn't completely crush in on you, some cars have ribs, that go around where the door frames are. And if you're wearing your seatbelt, if you run it right off in a bank and your car rolls down, all right, that's what you want coming from 60 miles an hour. Right. That'll do it. Just stretch it out. Stretch it out. Stretch it out. Okay. So I've heard of police who've come up to and they see how banged up the car is and use that.

as evidence that the person should have been hurt badly. When the more banged up the car is, provided the cavity where the driver is, is still intact, right? The rest of the car, let it get banged up. That's what you want in order to absorb all that kinetic energy. That's what I'm trying to say here. And by the way, that's why high jumpers and pole vaulters land in pillows. and not on cement. Well, yeah. If you come over a high jump or a pole vault and you fall down gracefully backwards and you hit cement, you just broke all

your ribs and your back. Okay. Because they actually instituted that about five jumps in, but yeah.

The five guys that didn't make it. So, you know, we should put some... Some should go under here, guys. Guys, I'm not going next. Some questions. Why did you think it took five times to figure it out? There's some physics idiots there, figure that one out. So, you want to stretch out the time. So, the deeper that cushion is, the less injury there is. And there's so little injury that they get back up and continue. All right? They don't have to, like... go to the hospital every time that happens. Yeah. That's like when they shoot the guy out of the cannon and they shoot him into cardboard boxes, empty cardboard boxes that have all been stacked on one another. Oh, okay. Yeah. And so each box just my guy got shot into a net. Did you go to the budget circus? The trash boxes. Listen, I went to a ghetto circus. Okay. That's what I'm saying.

It was very exciting. My people had nets. I don't know where you were. Go get some boxes. Because a net, of course, when you hit it, it's not rigid. It'll absorb your energy into the framework of the stitching. Yeah. But it also may throw you back out. Boxes will never do that. So what happens is, what that meant is it didn't completely absorb the energy. It takes the energy and it sends you back out the other way. Correct. So the energy bounces. That can happen. Okay. So, so anyhow, with a car, you want this, it's why head on collisions are so deadly because the car doesn't have a chance to slow down. Right. And the whole point of the airbag is to give you a chance to slow down in a very abrupt front end collision. All right. So the airbag will get there before you will because it expands explosively. It moves at the speed of sound basically when it inflates and that's faster than it would take you to actually hit the dashboard because your car is not going at the speed of sound in the other direction, right? So the bag beats you to the collision you would have with the windshield or with the dashboard and so and then it's basically a marshmallow. So that helps because it's soft and it's cushiony and it increases the time it takes you to come to a stop.

your body to come to a stop. So a safe collision is the one that takes the longest. Provided your driver space is not compromised. So when they say it's a slow motion car crash, that's actually a good thing. It's a good thing.

That's a good thing. So that's all. That's all I try to say here. Well, I think the lesson here that we've all learned is take the train. Take the train. That's the real lesson here. The train is the way to go. Because even if you survive it, you really don't want to be on that highlight reel is what you're saying. Right. That's not the thing. All right. So here's something to look for. Maybe we could put this in a, in the sports edition. If you ever want to know how badly hurt the batter is in baseball, if they get hit by a pitch, right? Look to see how much the direction of the ball changed after it hit them. Okay.

Right. So if it hits their thigh, but keeps going and gets caught in the catcher's mitt, not a problem. Right. Okay. It'll sting, but it's not. Okay. If it hits a thigh, bounces forward. Oh my gosh. Cause that meant all the kinetic energy of the ball got absorbed by the muscle and the bone and got sent back. So watch it now that we have it all in slow mo, high speed cameras, you can see what the exact trajectory of the ball is. Right. And the more the ball continues to roll back to the backstop or go to the side or go forward is the full information you need about how much damage it would have caused the batter. And if you don't see that, you know that how much it hurt when the batter rushes the mound to beat the hell out of the picture. That's what you know, we really hurt. Plus there's another thing with bat with baseball, you know, it's not a violent sport.

Yet if they get hit by pitch, they're not going to rub it, right? No, they're tough. It's on. They don't want to show the pitcher that the pitcher hurt him. They walk it off, as they say. They walk it off, exactly. And then I didn't even go rub the spot. They got too much pride. So imagine if they were World Cup soccer players. Oh, my God. If you hit a soccer player with a baseball, he would just fall down. I'm shocked! They always hold their knee or their ankle, right? Even no matter what. The Academy Awards for that. Anyhow, that's all the time we got Chuck. Alrighty. So like I said, anytime you see one of these accidents, if someone says it was a miracle, no, it's not, it's just physics.

It would have been a miracle had Dave Dale Earnhardt survived his crash. Oh, and one other thing, there's a record for the person who survived the highest fall from a building. All right, now there's a height above which basically 100% of people are dead. This person fell from a higher fight and

survived. And they say, the person was religious and Jesus saved them or whatever. But if you, okay, I heard this and I said, I need more data here. And then I looked at the building, the person hit a tree on the way down, okay? And broke branches of the tree. Oh, that's a good thing, because the kinetic energy of the body is now going somewhere else, rather than breaking his own bones, and probably broke his food bones nonetheless. But in a fall that would normally kill you, if a tree is there, it's gonna help you not die. Anything that's in the way is gonna help you not die. So all right, Chuck, that's all we got. What kind of car accident to have?

to maximize your chance of surviving it. And don't ever run full speed ahead into a wall, please. Too late for that. All right, I'm sorry I shouldn't have made that joke but it was for a minute that was a Slogan that meant that they wanted to drill for oil and stuff like that. All right drill baby drill, but sorry I Apologize and anybody out there. I don't mean anything buy it. Okay. Yes. Go ahead. All right So burn baby burn just as a as a concept I don't if you ever thought much about what it is for something to burn have have you It's a disco party I want to be at every time. That's what it is. Burn, baby, burn. Disco and burn. Oh, burn, baby. Burn that, mother. Yeah. All right. So here's the thing. I have never thought really about what it takes for something to burn, except that I know that anything that outside the presence of oxygen won't burn. OK. So burning just in common usage of the term.

means combustion in the presence of oxygen. That's what we generally mean when we say something is burning. So that's why, for example, you might have done this experiment in maybe middle school or possibly as early as elementary school. You could take a piece of paper, light it, but then put some kind of dome over it. And then it eats up all the oxygen inside and then just extinguishes itself. So it needs the oxygen to burn. All right. Now, I got a question for you. Have you ever burned water? OK. The answer is no. Just don't stop thinking. No, I want to hear a no. Just wait a second. Just wait a second. Because I know you're an astrophysicist, but maybe I should explain a couple of things to you now. Of course, I've never burned water. Because water doesn't burn.

If you can imagine taking a cube of pure aluminum, okay? I cannot, but go ahead. Okay, just imagine a chunk of aluminum and putting it on the stove. Put it in the oven. Okay. Okay. Will it burn? Aluminum in a jar. I don't know if it'll burn. No, it just gets hotter, right? Yeah, it gets hot. That's why we can make pans out of aluminum, okay? because they don't burn. Okay, so the point is, notice, you'll only use the word burn for some things and not others. Correct. In fact, the thing that's on fire. The things you're most likely to use the word burn for are the things that turn black after they have burned. True. That is true. Basically, most things that burn, you know, they're black afterwards. Except water.

except aluminum. Except I can go down a list, okay? Except lead. Except I can go down the list. There are plenty of things that can burn that, sorry, that can get hot, not turn black, and we would never say they were burning. True. Okay, so the question then is- That is true. My cast iron pan, it gets hot as hell. It's sitting on fire, actual flame. Flames can be on it, and it's not burning. And it never burns. Okay. Okay, so what do all the things that burn have in common? All the things that burn would they, I don't know, they catch fire? I don't know. What they have in common is you park the curtains, molecularly park the curtains, and you find the carbon atom. I was gonna say carbon, but isn't carbon in pretty much everything? It's in everything you will ever eat, except one thing.

Except two things salt and water I knew the salt and I knew the water. Yeah, you ever burn salt. No, you can't okay. This is my point Okay, by the way salt is crystals sugar is crystals. You can burn sugar Yeah, okay, and sugar does it catches fire and it melts and it melts catches fire burns turn black the whole thing whole deal so so other than salt and water everything else you eat Contains the carbon atom deep down within the molecules right a be the molecules that contain carbon atom have in themselves chemical energy Correct. Yes. Yeah, that's why you eat food at all first place because you take get the energy from the food So your body can use the energy. Be it a dead plant or a dead animal, I don't care. You're eating it to get in large measure, there's some nutrients here after two, but in large measure, you get in your calories to get the energy that they have. Okay, so now watch. So you have these molecules that have chemical energy. When you burn it,

you're breaking these chemical bonds with the help of oxygen. Right. And the act of breaking these bonds releases energy.

Your body does that but if you do it with a flame all you have to do is start it and Like a corner of a piece of paper and the rest of the paper burns Because a little bit that you started it breaks the chemical bonds that are sitting above it Which causes a chain reaction a chain reaction all the way up all the way on up and so When you and by the way wood campfire the wood Park the curtains, there's the carbon atom in there. Right. And wood contains all the energy that it had pulled out of the environment to become a tree in the first place. That's right. Then you cut it down. Oh, by the way, what kind of wood burns best? Well, it's got to be dry. Dry wood. Why? Because you can't burn water. Right. And the worst thing in the world is to have dry wood with a wet spot inside of it. And then while it's burning, It says, hey. I didn't know what's talking to me. It's a word. It's speaking English. It starts speaking, man. That's how you know it. Exactly. Man, what's wrong with you? What's wrong? Right, because a little pocket of water, it's not going to burn. All it will do is get hotter, become steam, expand, and then pop. Right. Pop.

So that's why you want to dry out the wood, get all the water out of there so that it's just these fat carbon molecules waiting for you to extract the energy from them and make the beautiful fire in the fireplace. And so depending on the chemical reactions, depending on how efficient or inefficient they are, you can leave behind layers of carbon. And that's what we basically call soot. Right. Soot. By the way, it's possible to burn something so there's no soot left over because then the carbon gets uptook into other molecules that are stable and float away like CO2. Okay, so carbon dioxide is a product of burning wood and other things and that's not carbon left that turns black. That's just a whole other molecule that's transparent and you can't smell it. You can't taste it and it's in the atmosphere. But if you have certain kinds of combustion, certain kinds of burning that don't successfully turn it all into gas, you will get a residue at the bottom, and there is carbon ash. And it's carbon. You're revealing the carbon out of which all of life is based. Look at that. That's pretty doggone cool. So whenever you have a fire, it's basically the circle of life. That's exactly. So you should hold that wood up too. Before you burn it.

Whatever the hell he said. And then you burn it. Then you burn it and you pay due respect for the tree that gave its life, for the molecular energy contained in it and the fact that life is carbon-based. Yes. All of you burn it. Actually, you know what? That is very poetic, actually. It's a beautiful thing. It's a beautiful thought. I love it. And... The CO2 you just put into the air, that's not contributing to global warming because the tree took it out in the first place. So you are carbon neutral in the process. You are carbon neutral provided you're not only just cutting down trees to feed your fireplace. You got to make sure the trees planted somewhere and you're basically carbon neutral, carbon dioxide neutral in that exercise. So there you have it, kumbaya around the fireplace. I love it. I love it.

I'm gonna go burn something down right now. Shut nice up on arson charges. I was just trying to show people the beauty of carbon, man. I just think about it because we always, historically, made houses out of wood. Right. And wood is basically fuel. Building your house out of fuel and you're surprised it burns down. Right. Cavemen never had house fires. No, never. And the Brick House, the pig that lives in the Brick House, nobody's burning down his house. Right, exactly. OK. That's super cool, man. All right, we got to call it quits there. All right. Chuck, always good to have you. Always a pleasure. Neil deGrasse Tyson, as always, keep looking up.